Nathaniel Recto

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Education

Bachelor of Engineering, Computer Engineering Sep. 2022 – Jun 2026
Toronto Metropolitan University

Technical Skills

Programming: Python, C/C++, Java, Swift, Kotlin, SQL, VHDL.
Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, MobileNetV2, YOLOv8, fine-tuning.
Evaluation: Stratified cross-validation, hyperparameter tuning, probability calibration, ROC-AUC, F2, Brier score, failure-mode analysis.
Model Deployment and Software: TensorFlow Lite, Flask, SQLite/SQLAlchemy, SwiftUI, Core ML, Git/GitHub, Linux/Unix, automated testing.
Data and Computer Vision: OpenCV, MediaPipe, Pandas, NumPy, Matplotlib, data curation, object tracking.

Projects

Basketball Shot Tracker (iPhone App) | *Python, PyTorch, YOLOv8, OpenCV, NumPy, Swift* Oct 2026

- Built a computer-vision system that calls basketball shots made or missed from one phone camera, pairing a fine-tuned **YOLOv8s** detector (**8,347** images, **0.85** val mAP50) with a physics-based tracker and shot judge.
- Hand-labelled **555** shots blind from **16** public videos and **11** own sessions in **2** gyms, splitting dev and test by whole session before filming to prevent data leakage, and scoring the test set once with frozen, fingerprinted code.
- Achieved **85.6%** end-to-end accuracy (**95%** CI **77**–**91%**) on **90** held-out test shots in an unseen gym, finding **87** of **90** shots with none invented, after detector retraining lifted own-footage accuracy from **13.4%** to **78.2%**.
- Deployed it as a native **Swift/SwiftUI** iPhone app running **Core ML** on the Neural Engine at **28**–**32** fps (**45**–**62** ms per call), verified by **132** automated tests to match **Python**'s **123** shot calls exactly across **33,386** frames.

Heart Disease Risk Prediction System | *Python, scikit-learn, XGBoost, Pandas, Flask, SQLite* Oct 2026

- Built an ML pipeline for heart disease screening on **918** UCI patient records from **4** hospitals, using a stratified **80/20** split (**734** train, **184** test) and leakage-safe scikit-learn Pipelines for imputation, scaling, and encoding.
- Evaluated Logistic Regression, Random Forest, and XGBoost using stratified **5**-fold cross-validation and RandomizedSearchCV to optimize F2 for recall, selecting XGBoost with the strongest cross-validated F2 of **0.85**.
- Tuned decision thresholds and calibrated probabilities on out-of-fold predictions, reaching **99%** recall (**101** of **102** sick patients), **0.90** ROC-AUC, and a **0.125** Brier score on the held-out test set, cutting missed cases from **13** to **1**.
- Served the models through a Flask web app with a **4**-step assessment, **3** selectable models, per-prediction explanations, and private SQLite/SQLAlchemy history, validated by **7** automated tests.

Real-Time Driver Drowsiness Detection System | *Kotlin, Python, TFLite, MediaPipe* Sep. 2025 – Apr 2026

- Integrated 2 TensorFlow Lite models in a Kotlin app with CameraX and MediaPipe, enabling real-time driver drowsiness detection through on-device ML inference.
- Curated a four-class eye and mouth dataset, then trained and fine-tuned both MobileNetV2 TensorFlow Lite ML models, achieving **99.17%** eye-state accuracy and **96.18%** yawn-detection accuracy.
- Implemented a scoring algorithm combining 4 drowsiness signals—eye closure, yawn, head tilt, and model output to trigger driver warnings when the combined score reached the 100-point threshold.
- Evaluated model outputs across lighting, head angles, facial movements, and combined drowsiness scenarios, identifying failure modes and defining operational limits for on-device inference.

Experience

Operations Support May 2024 – Aug 2024, May 2025 – Aug 2025
Enercare *Markham, ON*

- Resolved HVAC contract problems by collaborating with cross-functional teams in energy advisory, sales, management, and sales support, helping maintain installation and service operations.
- Processed contractor payments, invoices, and work orders in GERS, Clarify, and MS Office, reducing processing time by 20% through accurate record updates.
- Reduced billing documentation errors by reviewing rental extras, district extras, equipment exchanges, installs, and removals and updating contract records.
- Coordinated administrative requests and escalations as the key team contact, using tracked follow-ups to increase business-support productivity by 15%.